

C++2: Lesson 2 – Lab Project

1. Add **summary** at the top and **comments** where necessary.
 - Recordings and interviews are an exception, but they should be included in your submission code.
2. **Cite** the source of your information.
3. Refrain from including anything **you don't fully understand**.

2 100 points Use a class named Table to: Prompt the user to input the length and width of a table (in feet). Calculate the surface area using: Surface Area=Length × Width Compute: Cost= Surface Area × 15 Price= Surface Area × 25 Profit= Price–Cost Requirements: • Implement the Table class with private member variables for the table's dimensions and methods for calculations. • Include methods to input dimensions and display calculations. • Place both the Table class and the main() function in a single file for simplicity <table> <tr> <td>Sample output</td> <td> Enter the length of the table (in feet): 5 Enter the width of the table (in feet): 3 Surface Area: 15 square feet Cost: \$225.00 Customer Price: \$375.00 Profit: \$150.00 </td> </tr> <tr> <td>File(s) to submit</td> <td> C202_2.cpp C202_2.png </td> </tr> </table>	Sample output	Enter the length of the table (in feet): 5 Enter the width of the table (in feet): 3 Surface Area: 15 square feet Cost: \$225.00 Customer Price: \$375.00 Profit: \$150.00	File(s) to submit	C202_2.cpp C202_2.png	
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3 100 points Write a class named Circle that includes the following: 1. Variables: -radius: A private member variable to store the circle's radius. -pi: A private constant initialized to 3.14159. 2. Functions: - default constructor that initializes radius to 0.0. - setter function to update the radius value. - getter function to return the circle's radius. - function to calculate and return the area of the circle using the formula: Area = pi × radius × radius - function to calculate and return the diameter of the circle using the formula: Diameter = 2 × radius - function to calculate and return the circumference of the circle using the formula: Circumference = 2 × pi × radius					

3. Program Requirements: - Prompt the user to input the radius of the circle. - Create an object of the Circle class and set its radius using the user input. - Display the calculated area, diameter, and circumference of the circle. - Place both the Table class and the main() function in a single file for simplicity	
Sample output	Radius? 5 Radius: 5.00 Area: 78.54 square units Diameter: 10.00 units Circumference: 31.42 units
File(s) to submit	C202_3.cpp C202_3.png